OOS Impact Analytics — Methodology Sheet

How the dashboard detects out-of-stock and demand-throttling events and values their revenue / contribution-margin impact. Worked example: **VV-VITA-311 — NAC Capsules** (amazon.it). Prepared for review.

1. Data sources

• **Amazon FBA Inventory Ledger** — real daily sellable warehouse balance and units shipped per SKU. The account runs **Pan-EU**, so stock is pooled: a SKU is physically out only when the whole EU balance hits zero.
• **Novadata margin export** — daily units, sales and contribution margin (CM3) per SKU and marketplace; supplies price and CM3 per unit for valuation.
• **Shopify catalog** — English product titles.

2. Two core measures

Demand rate (lambda). Trailing 90-day average units per calendar day, per SKU and marketplace. Daily sales follow an approximate Poisson process, so the chance of selling zero on a normal day is P(0)=e^(-lambda): ~37% at 1/day, ~5% at 3/day, ~0.7% at 5/day. A zero-sales day is therefore only treated as a stock-out when lambda is high enough that zero is a real anomaly (default threshold 3/day).
Reach (days-of-supply). EU sellable stock divided by trailing 28-day average units shipped — how many days of cover remain.

3. Daily classification (one label per SKU x marketplace x day)

Evaluated in priority order; the first match wins:

Priority	Label	Condition	Counts as
1	Physical OOS	EU sellable balance = 0	Lost (involuntary)
2	Critically low	Reach < 3 days	Lost (involuntary)
3	Cooling down	3 <= reach < 30 days, still selling (units > 0), throttled (price >= +8% and/or ad cut >= 50%)	Miss (voluntary)
4	Marketplace gap	Units = 0 on a day enclosed by sales, lambda >= 3/day (incl. throttles that hit 0)	Lost (involuntary)

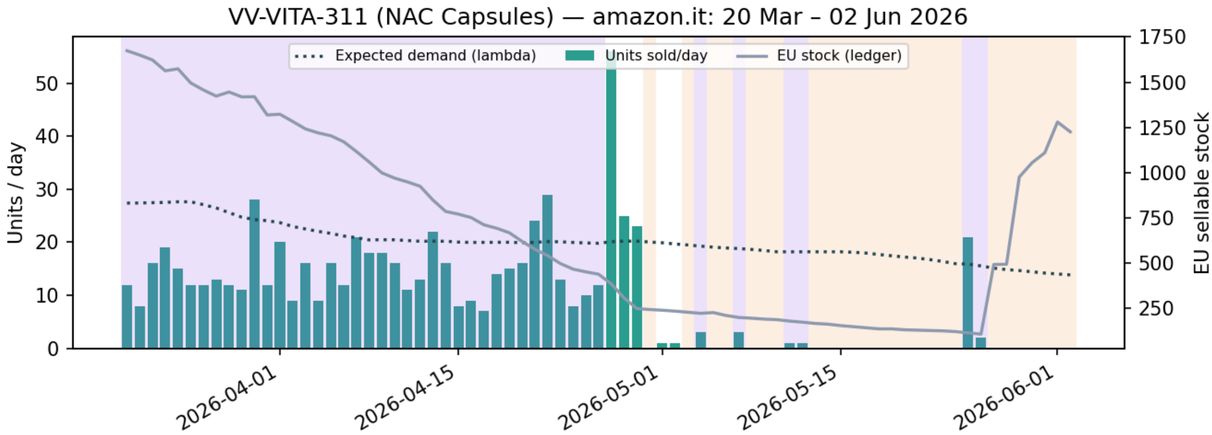
Guards: the SKU must already be selling (ignores pre-launch); a zero-run must be enclosed by later sales or be within the last 21 days (ignores discontinued tails). Cooling-down separates deliberate throttling — which otherwise looks like a demand drop — from genuine stock-outs, so the two are never double-counted.

4. Impact valuation

On any flagged day: **missed units = max(lambda - actual units, 0)** (the expected demand we didn't capture). Valued at the SKU's trailing average selling price and CM3 per unit:
Lost / Miss revenue = missed units x avg price
Lost / Miss CM3 = missed units x avg CM3 per unit (the true P&L impact).
'Lost' = involuntary (OOS); 'Miss' = voluntary (cooling down). Consecutive flagged days are grouped into discrete events.

5. Worked example

VV-VITA-311 — NAC Capsules, amazon.it. Full year on this marketplace: **32 OOS days** (lost €10.434 revenue / €1.379 CM3) and **56 cooling-down days** (miss €9.301 revenue / €1.534 CM3).



Highlighted window 20 Mar 2026 – 02 Jun 2026: EU stock ranged from ~1.674 down to ~108 units and reach bottomed at ~6 day(s). In this window the model flagged **26 OOS day(s)** (red/orange shading) and **44 cooling-down day(s)** (purple) — the SKU was throttled (price up / ad spend cut) while stock was tight, then went into genuine stock-out as reach collapsed. Note the balance rarely hits literally zero, so the **reach < 3** rule, not a balance-of-zero rule, is what catches the hard stock-out.

What "cooling down" means here. On the purple days the SKU kept selling (units > 0) but demand was deliberately damped to stretch the remaining stock to the next shipment. **Before cooling down** it sold at ~€18.26 with ~€48/day of ad spend; **during the throttle** the price was lifted to ~€19.99 and ad spend cut to ~€28/day. Sales fell to about 12/day versus the expected ~20/day, and that gap is booked as a **voluntary miss** — a pricing / advertising choice to protect availability and the listing's ranking — not a lost stock-out. Crucially the SKU keeps selling: the moment a throttle pushes sales to zero it is no longer cooling down and counts as OOS (lost). Cooling down trades a little volume now (the miss) to avoid the larger ranking and lost-sales hit of a full stock-out.

Representative days (Lost/Miss shown as CM3 €):

Date	Units	λ	EU stk	Reach	Price	Category	Lost €	Miss €
03-20	12	27	1674	30	19.69	Cooling down		43
03-21	8	27	1650	29	19.99	Cooling down		54
03-22	16	27	1623	29	19.88	Cooling down		32
03-23	19	28	1563	28	19.99	Cooling down		24
03-24	15	28	1574	28	19.99	Cooling down		35
03-25	12	28	1496	27	19.99	Cooling down		44
03-26	12	27	1457	27	19.84	Cooling down		43
03-27	13	26	1423	26	19.99	Cooling down		38
04-27	56	20	385	10	16.93	—		
04-30	0	20	244	7	–	Marketplace gap	45	
05-03	0	19	229	7	–	Marketplace gap	41	
05-05	0	19	227	7	–	Marketplace gap	39	
05-06	0	19	211	7	–	Marketplace gap	38	
05-08	0	19	196	7	–	Marketplace gap	36	
05-09	0	18	191	7	–	Marketplace gap	35	

6. Caveats

Impact figures are estimates (assume the SKU would have sold at its recent rate). Ad-spend-cut detection only applies from when Novadata began reporting Advertising Costs (~Feb 2026); the price-hike lever spans the full year. Reach is EU-pooled, so reach-based rules need ledger coverage for the SKU. All thresholds (3/day, reach 3 & 30, +8%, -50%) are tunable in the dashboard.